Korea Technology and Information Promotion Agency for SMEs

*(max 10 pages)*

# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information

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## 2. Technology Overview

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| **Title of Proposed Technology** | *e.g. A technology that allows maintaining temperature fluctuations within a range of ±3℃.*<br>RETRO-CHIP: Deployable Methods for Chip-Level Replacement and Repackaging to Retrofit and Upgrade Legacy Electronic Systems | | | |
| **Korea's 12 National Strategic Technology Fields** | Semiconductor/Display **(√),** Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine **( )**, Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing **( ),** Quantum ( ) | | | |
| **Industrial Technology Classification** [1] | **Technology Field 1** (Large Category) | | **Technology Field 2** (Large Category) | |
| | Electrical・Electronics | | Electrical・Electronics | |
| | Medium Category | Small Category | Medium Category | Small Category |
| | Semiconductor Devices and Systems | Semiconductor Devices | Semiconductor Equipment | Packaging Equipment |
| **Concept of Proposed Technology** | *Describe the core concept, main functions, and potential uses of the proposed technology*<br>Legacy electronic systems in industry and ICT increasingly face component obsolescence, where critical chips, in particular complex ICs, are no longer available, forcing costly redesigns and their validation or premature replacement of otherwise functional equipment. This project is motivated to methodologies and technologies to provide an economical and sustainable retrofit approach for obsolete chips. The primary objective of the project is to utilize the existing pin-footprint of the old chip on the printed circuit board and create a behavior-compatible new chip package with alternative | | | |

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| | electronic components. The benefit of this project is to prolong the lifetime of premium electronic devices in e.g. telecommunication, medical, robotic, avionic, and defense applications. This approach will ensure the long-term availability and resilience of mission-critical equipment and industrial systems in various applications. It allows furthermore to update and extend functionality of the device to the state of the art (e.g. wireless interfaces or security options). This methodology combines resilience of well approved electronic devices with new possibilities of system connectivity and control. Lastly, the project will strengthen South Korea and Germany as self-sufficient industrial location under currently changing geopolitical conditions. |
| **Rationale for Joint R&D Project** | *Describe the need for this joint R&D project and why collaboration is required*<br>German-Korean cooperation is not merely an advantageous addition to the Retro-CHIP project; it is a strategically viable option. Due to the fact, that it combines complementary industrial strengths that are not consistently available in Germany to the same extent or on the same scale. Furthermore, it serves to reduce the current growing supply chain risks in the global electronic industry and explicitly address economic and environmental advantages of electronic repair and upgrade. A German–Korean collaboration is needed, because it combines the complementary strengths that are critical for achieving the retrofit objectives. Especially where legacy electronics must be upgraded under obsolescence, while preserving function, interfaces, reliability, and manufacturability. |
| **Key Words** | 1. Component obsolescence mitigation<br>2. Chip replacement<br>3. Customized package and interposer design<br>4. Selective de-soldering and disassembly process<br>5. Techno, economic, and environmental feasibility assessment |

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## 3. Research and Development Objectives & Plan

| Technology Readiness Level (TRL)[2] | Start of the Project | Level (3) | End of the Project | Level (5) |
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| R&D Objectives | *Describe the specific technologies you aim to develop through this R&D project*<br>The overall research objective is to develop and validate a practically deployable, science-based methodology for retrofitting and upgrading legacy electronic systems under component obsolescence.<br>The project aims to enable the replacement of unavailable chips by:<br>**1.** Preserving functional behavior and external interfaces.<br>**2.** Ensuring reliability after disassembly and reassembly.<br>**3.** Providing a structured decision process to select the most suitable retrofit pathway (chip-level replacement, functional block replication, or software-based emulation).<br><br>The outcome will be a replicable end-to-end retrofit approach, from circuit assessment and verification through package/module redesign and manufacturing process qualification to an AI-based techno-economic and environmental assessment. | | | |
| R&D Target Specifications | *Describe the target performance specifications to be achieved through this R&D project, including quantitative values such as measurable parameters*<br>• Development of a method for the systematic analysis of chip functionality and design that needs to be replaced.<br>• Development of a system model for the validation of the required functional specification with available packaging and interposer technologies.<br>• Development of a realistic package behavior model for assessing the electrical, signal, and thermo-mechanical system requirements.<br>• Computer-aided, numerical optimization of the electrical, signal and thermo-mechanical integrity with respect to system reliability. | | | |
| R&D Plan | *Provide detailed plan outlining the research approach, key tasks, methodologies, expected milestones, and the timeline required to achieve the stated R&D objectives*<br>**WP1 (IEM/IZM): Circuitry Analysis and Retrofit Capability Assessment**<br>WP1 establishes the technical foundation for retrofitting by analyzing the selected electronic systems with respect to circuit design, functional behavior, and retrofit feasibility. The work develops a theory-driven and methodical assessment approach to determine whether and how obsolete and discontinued components can be replaced while preserving function, interfaces, and reliability. Representative product examples will be used to derive generalized assessment criteria and decision rules.<br>Key tasks include:<br>**1.** **Analysis** of the obsolete **chip functions** and **specification** of **requirements** for the retrofit solution.<br>**2.** **Specification** of a **testing** and **validation** concept with particular focus on **reliability requirements**.<br>**3.** **Selection** and **evaluation** of optimal **de-soldering** and **disassembly concepts**.<br>**4.** **Definition** of **technical requirements** for replacement and **re-soldering** (including constraints on pads, materials, and interfaces).<br><br>**Deliverable:** Assessment strategy and associated requirements for replicating chip functionality and form factor.<br><br>**WP2 (IEM/IZM): Retrofit Options and Hierarchical Implementation Strategy**<br>WP2 develops a structured, hierarchical implementation strategy for retrofitting and upgrading legacy electronics under component obsolescence. The strategy explicitly covers multiple technical pathways and provides criteria for selecting the most appropriate solution depending on performance, risk, cost, and manufacturability.<br>The following retrofit options are addressed: | | | |

1. **Rerouting with identical chip functionality**, including adaptation of interface behavior (electrical and signal-level).
2. **Functional replication at block level**, enabling multi-component replacement solutions where a single obsolete device is replaced by an equivalent sub-system.
3. **System behavior emulation**, shifting functionality into a software-based solution where feasible, while maintaining external behavior and compliance requirements.

**Deliverable:** Hierarchical retrofit option strategy with selection criteria and implementation guidance.

**WP3 (IZM/IEM): Design of Alternative Chip System / Module with Focus on Package Design**
WP3 designs the alternative chip system or module required for the selected retrofit pathway, with emphasis on packaging and system integration. A model-based design concept will be developed to describe the system architecture and ensure functional equivalence and interface compliance.
Key tasks include:

1. **Development** of the **system architecture** concept and **system model**.
2. **Design** of the **electrical circuit** and finalization of the **wiring diagram**
3. **Simulation of critical signal and power paths to verify integrity and robustness.**
4. **Derivation of appropriate component technology and system design parameters (e.g., build-up layers, routing concepts, and integration options).**
5. **Simulation of the final system model to analyze functional behavior and verify that retrofit objectives.**

**Deliverable:** Alternative module design concept, validated system model, and simulation-based verification results.

**WP4 (IZM/IEM): Layout and Packaging Concept Including Thermo-Mechanical Requirements**
WP4 translates the selected design concept into a manufacturable layout and packaging solution, explicitly considering thermo-mechanical constraints and electro-mechanical integration. The aim is to ensure that the retrofit solution can be produced reliably and integrated into existing assemblies without compromising performance or lifetime.
Key tasks include:

1. **Development** and **application** of a **morphological matrix** linking electro-mechanical requirements with **standard package technologies** to identify best-fit packaging options.
2. **Definition** of **design rules** for component placement and **packaging technology** selection.
3. **Adaptation** and **specification** of **manufacturing technologies** required for the custom design.
4. Layout of the interposer and the connection interfaces between package and printed circuit board

**Deliverable:** Packaging and layout concept including design rules, manufacturing specification, and thermo-mechanical compliance requirements.

**WP5 (IEM/IZM): Replacement Technology—Disassembly and Assembly Process Including Solder Pad Treatment**
WP5 develops the practical replacement technology needed to remove obsolete components and assemble the retrofit solution, including complete treatment and, if required, refurbishment of old solder pads. The work ensures process reliability and compatibility with legacy assemblies.
Key tasks include:

1. **Inspection** and assessment of **spatial boundary conditions** for component replacement.
2. **Selection** and **definition** of **re-soldering** technologies and processes.
3. **Post-treatment methods** including cleaning and optional **refurbishment** of **solder pads**.
4. **Adaptation** and **specification** of **soldering technology** to ensure repeatability, yield, and reliability.

**Deliverable:** Validated replacement process chain (disassembly → pad treatment → assembly) and process specification.

**WP6 (IZM/IEM): Economic Feasibility, Efficiency, and Environmental Benefit Assessment**
WP6 evaluates the retrofit approach from an economic and sustainability perspective, ensuring practical applicability and scalability in relevant industrial contexts. The work quantifies cost drivers, identifies viable business cases, and assesses the environmental impact of retrofitting compared to replacement procurement or complete system redesign. The impact on the supply chain in particular

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| | with regard to the possible annexation of relevant manufacturing countries (such as Taiwan) is of interest here, as this could have a negative effect on the prices and availability of high-quality semiconductors such as CPUs, GPUs and memory.<br>Key tasks include:<br>**1.** An **AI-based** study to **identify** profitable and **realistic business cases** (with emphasis on B2B equipment and ICT infrastructure),<br>**2.** **Development** of a **method** for realistic and up-to-date **component cost estimation** including **availability** and **supply chain** constraints (e.g., harvesting from used products),<br>**3.** **Process** and **material cost analysis** based on lifecycle inventory data,<br>**4.** **Integrated cost** and **carbon footprint analysis**.<br>**5.** **Definition** and **assessment** of possible **critical geopolitical scenarios**, particularly regarding **resilience** and **secure industrial capabilities** in Korea and Europe.<br><br>**Deliverable:** Techno-economic and environmental assessment report including business case evaluation and resilience impact analysis. |

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## 4. R&R (Roles and Responsibilities)

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| **Expected R&R of Research Institute** | *Describe the expected roles and responsibilities of the research institute, such as technical contributions, research scope, resources to be provided, and specific tasks to be provided, etc.*<br>Applying German Fraunhofer Institutes:<br>• Fraunhofer IEM: system requirements and architecture, modeling and simulation of system behavior, de- and re-soldering technology<br>• Fraunhofer IZM: Package and interposer design, thermal and mechanical system behavior and simulation, cost and environmental assessment<br>Possible Korea research institute partners with focus on electronic packaging:<br>• KAIST (Advanced Packaging & Thin Film Lab),<br>• KETI (advanced/system packages and testing) |
| **Expected R&R of Korean SME Partner** | *Describe the expected roles and responsibilities of Korean SME, such as technical contributions, development tasks, testing/validation activities, resource commitments, and collaborations efforts, etc.*<br>• OEMs or tier 1 for providing industrial/systemically relevant driven use-cases<br>• Chip and/or electronic package design for developing the retrofit chip<br>• IC manufacturer for providing bare dies<br>• Electronic assembly with the task of de-soldering and re-soldering<br>• Electrical testing and optical inspection for rework<br>• Electronic repair and refurbishment specialist<br>• Electronic module and ICT system manufacturer for providing B2B hardware examples e.g. robotic, military, transport, medical, telecommunication |

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## 5. Expected Outcomes/Impact

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| **Expected Outcomes and Impact** | *Explain the expected outcomes of this joint R&D project, including technological achievements, commercialization potential, and broader industrial or economic impact, etc.*<br>• Comprehensive **technical assessment** framework for **chip/module retrofit.**<br>• **Design rules** and **workflow** for alternative **chip packages** (replacement package).<br>• **Modeling** and **optimization** of **retrofit chip** and **package design** including simulation.<br>• **Technology** for **de-soldering**, remaining **solder-pad refurbishment** and **re-soldering**<br>• **Reliability testing** and **technology verification**<br>• **Techno**, **economic** and **environmental assessment** of retrofit process |

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## ☐ Terms:

### 1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

### 2) Technology Readiness Level (TRL)

| Description | | |
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| ·A consistent and objective indicator of the technological maturity of key element technologies | | |
| | **Basic Research Stage (1-2)** | 1st Stage: Basic theories and experiments<br>2nd Stage: Proof of Concept, such as practical ideas or patents |
| | **Experimental Stage (3-4)** | 3rd Stage: Verification of basic performance at a laboratory scale<br>4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale |
| | **Prototype Stage (5-6)** | 5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems<br>6th Stage: Production and performance evaluation of prototypes at a pilot scale |
| | **Productization Stage (7-8)** | 7th Stage: Reliability testing and evaluation by potential customers<br>8th Stage: Prototype certification and standardization |
| | **Commercialization Stage (9)** | 9th Stage: Commercialization |